A PROJECT REPORT ON

BANK LOAN ANALYTICS

Data Analytics & Power BI Dashboard Project

Submitted by,

Hamsaveni P

Keerthana D M



Abstract

Bank Loan Analytics is a data analytics project that uses Microsoft Power BI Desktop to study a cleaned financial loan dataset of 38,576 loan records and 29 columns. The project summarises loan application volume, funded amount, amount received, repayment status, good-versus-bad loan classification, borrower characteristics, geographical patterns and financial risk indicators through a seven-page interactive dashboard.

The portfolio contains 33,243 Good Loans (86.18%) and 5,333 Bad Loans (13.82%). The total funded amount is approximately $435.8M, against approximately $473.1M received. Interest rates rise steadily from 7.35% for Grade A to 21.40% for Grade G, and California, New York and Texas lead in funded amount. The report documents the dataset, methodology, every dashboard page with its screenshot, the key findings, conclusions, recommendations and limitations.

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1. Introduction and Objectives

Lending institutions must balance growth in loan volume against the risk that borrowers fail to repay. Understanding which loans perform well, which are charged off, and how outcomes vary by grade, purpose, borrower profile and location helps teams monitor risk and make better-informed decisions. This project turns a loan-level dataset into an interactive Power BI report that makes these patterns easy to explore.

1.1 Project objectives

- Summarise the loan portfolio using key performance indicators (KPIs).
- Compare Good Loans and Bad Loans by count, amount, grade, purpose and state.
- Understand repayment status and the scale of charged-off loans.
- Explore borrower characteristics such as income, employment length, home ownership and verification status.
- Examine interest-rate and debt-to-income (DTI) patterns across loan grades.
- Build an interactive Power BI report with slicers and consistent visual formatting.

1.2 Tools used

Microsoft Power BI Desktop was used for data loading, modelling, DAX measures and dashboard design. The cleaned CSV file was imported directly using Get Data, so no live SQL or Python connection was required.

2. Dataset Overview and Data Quality

The analysis uses the cleaned file cleaned_financial_loan_data(1).csv, which holds loan-level records with borrower, loan, payment and date-related attributes.

2.1 Main fields used

Field	Purpose in analysis
ID	Unique loan/application identifier used to count applications
LOAN_STATUS	Repayment outcome: Fully Paid, Charged Off or Current
LOAN_CATEGORY	Good Loan / Bad Loan classification for portfolio comparison
LOAN_AMOUNT	Amount funded through the loan
TOTAL_PAYMENT	Total payment received
INT_RATE / DTI	Interest-rate and debt-to-income analysis
GRADE / SUB_GRADE	Risk-grade comparisons
PURPOSE	Loan purpose analysis
ADDRESS_STATE	Geographical analysis by state
ANNUAL_INCOME, EMP_LENGTH, HOME_OWNERSHIP	Borrower profile analysis
VERIFICATION_STATUS	Income verification analysis
ISSUE_DATE / YEAR / MONTH	Time-based loan application analysis

2.2 Data quality checks

- Dataset size: 38,576 rows and 29 columns; exact duplicate rows: 0.
- Missing ISSUE_DATE values: 4. These were left blank rather than assigned invented dates.
- Date fields are typed as Date, monetary fields as Decimal/Currency, and identifiers as text or whole numbers.
- DTI_PCT and INTEREST_RATE_PCT already hold percentage values, so their averages are divided by 100 before percentage formatting.

2.3 Loan classification

Fully Paid and Current records are grouped as Good Loans, while Charged Off records are grouped as Bad Loans. This is the classification stored in the dataset's LOAN_CATEGORY field.

3. Methodology and Measures

The project followed a six-step workflow:

- **Data loading:** Import the cleaned CSV into Power BI Desktop.
- **Data validation:** Check row count, column names, blanks, duplicates and data types in Power Query.
- **Measure creation:** Create DAX measures for counts, amounts, averages and percentages.
- **Exploratory analysis:** Compare status, grade, purpose, state, term, income, employment length, home ownership and verification.
- **Dashboard development:** Build seven report pages with slicers and page navigation.
- **Validation:** Compare KPI totals and category counts with the cleaned dataset.

4. Dashboard Design and Page-wise Description

The report has seven pages with a consistent dark background, gold highlights, white labels, page-navigation arrows and a Clear all slicers button. Each page carries slicers relevant to its analysis, such as month, purpose, loan status, state, term, home ownership and verification status. The screenshots below are taken from the Power BI report.

4.1 Page 1: Executive Summary

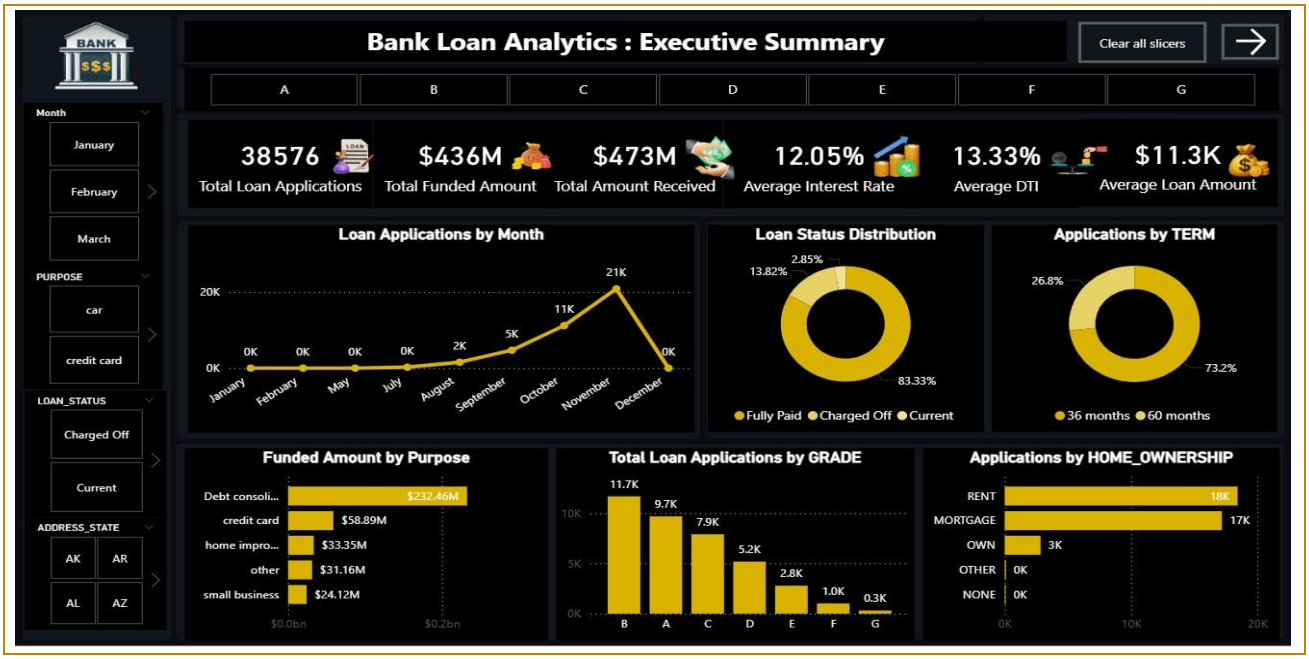


Figure 1: Executive Summary page

Gives a portfolio snapshot: 38,576 applications, $436M funded, $473M received, 12.05% average interest rate, 13.33% average DTI and $11.3K average loan. Supporting visuals show applications by month, loan status (83.33% Fully Paid, 13.82% Charged Off, 2.85% Current), term (73.2% 36-month), funded amount by purpose (debt consolidation $232.46M), grade (Grade B largest at 11.7K) and home ownership (Rent 18K, Mortgage 17K).

4.2 Page 2: Loan Quality Analysis

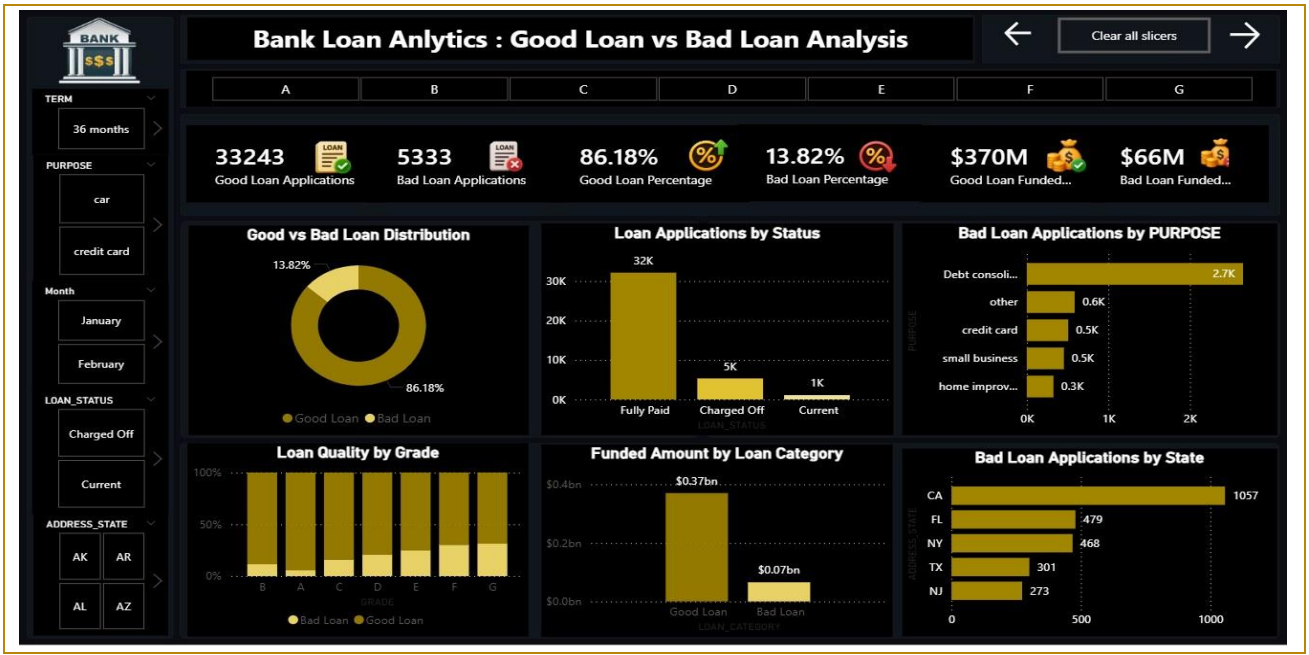


Figure 2: Good Loan vs Bad Loan Analysis page

Compares portfolio quality: 33,243 Good Loans (86.18%, $370M funded) against 5,333 Bad Loans (13.82%, $66M funded). The Bad Loan share grows with loan grade, and debt consolidation (2.7K) and California (1,057) have the most Bad Loan applications.

4.3 Page 3: Borrower Profile Analysis

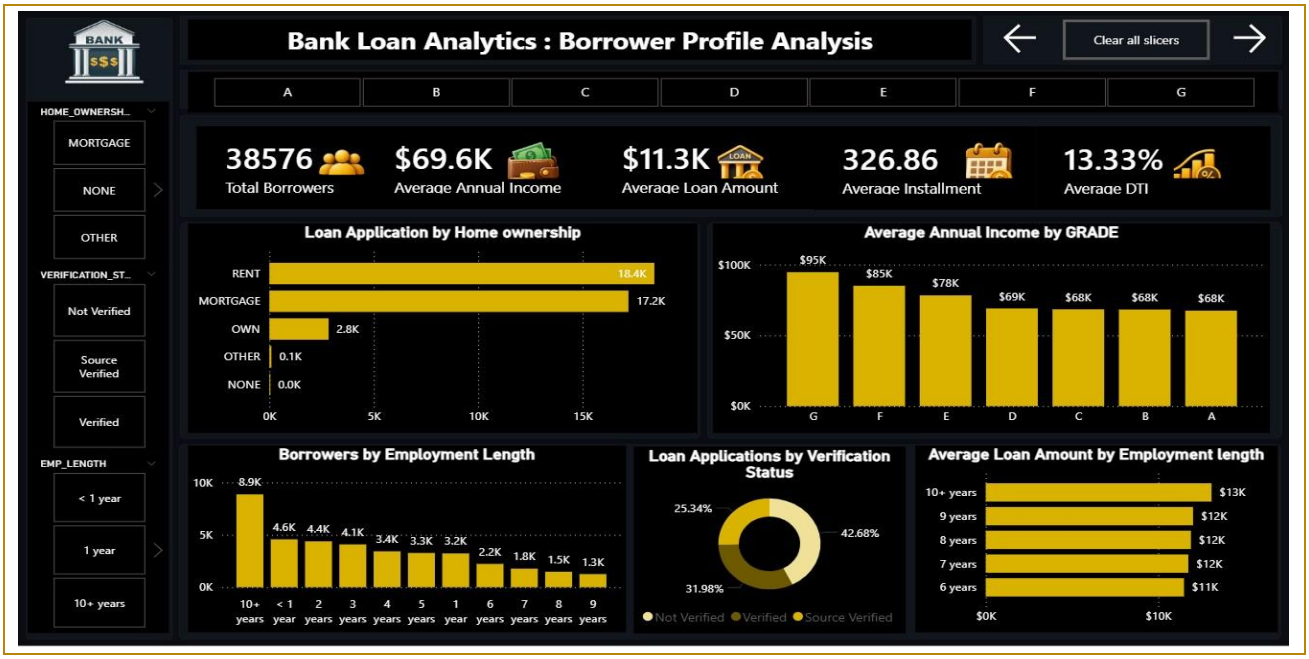


Figure 3: Borrower Profile Analysis page

Profiles the 38,576 borrowers: average annual income $69.6K, average loan $11.3K, average installment $326.86 and average DTI 13.33%. Renters (18.4K) and mortgage holders (17.2K) dominate, borrowers with 10+ years of employment form the largest group (8.9K) and also receive the largest average loan ($13K). Average income is highest for Grade G ($95K).

4.4 Page 4: Geographical Analysis

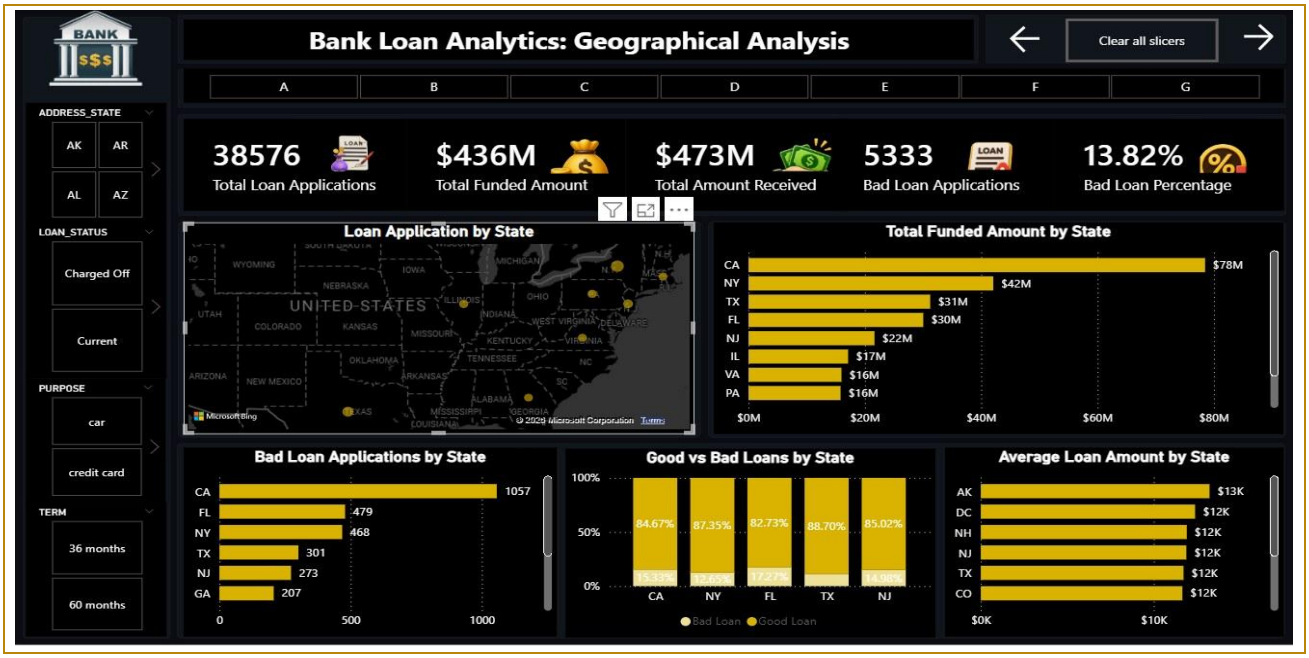


Figure 4: Geographical Analysis page

Maps applications by US state. California leads in funded amount ($78M), followed by New York ($42M), Texas ($31M) and Florida ($30M). California also has the most Bad Loans (1,057), and Florida has the highest Bad Loan share among the five states shown (17.27%).

4.5 Page 5: Financial & Risk Analysis



Figure 5: Financial & Risk Analysis page

Examines pricing and risk. Average interest rate climbs steadily from 7.35% (Grade A) to 21.40% (Grade G), while average DTI stays in a narrow 12.04% to 14.17% band. Scatter charts show loan amount against interest rate and DTI against annual income, and debt consolidation again leads charged-off loans (2.7K).

4.6 Page 6: Key Findings

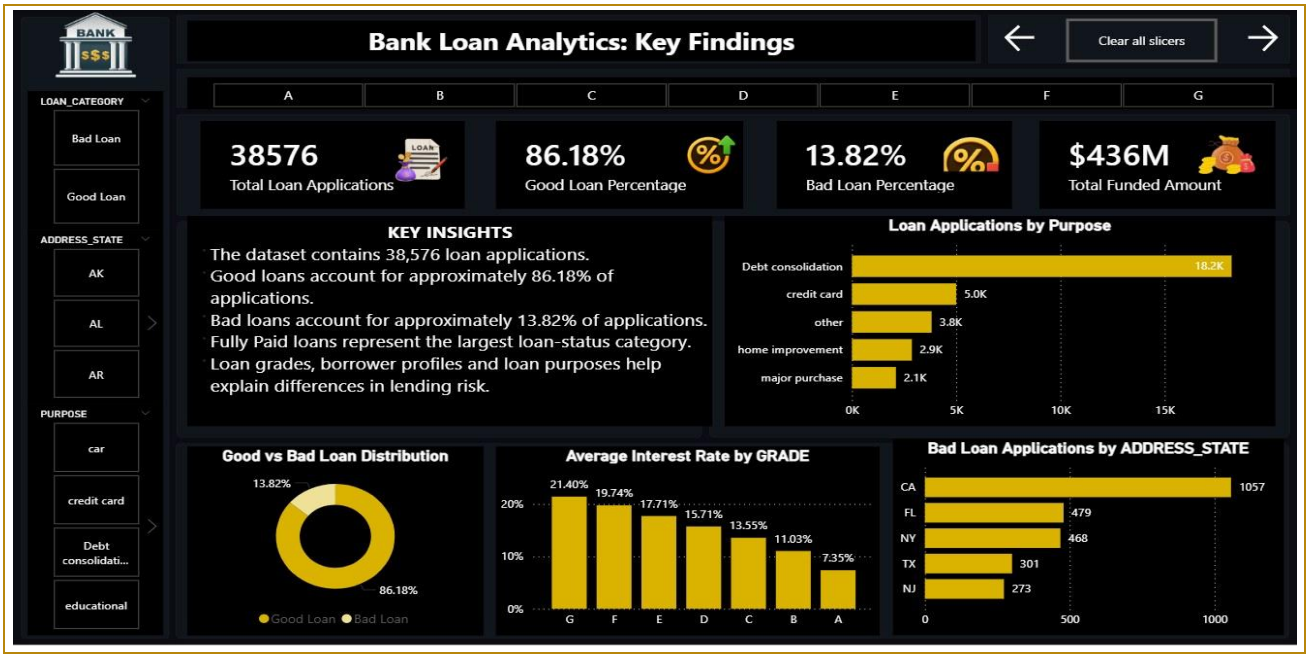


Figure 6: Key Findings page

Brings together the main insights with KPI cards, a written insight panel, loan applications by purpose (debt consolidation 18.2K), the good-versus-bad distribution, interest rate by grade and Bad Loan applications by state.

4.7 Page 7: Conclusion & Key Takeaways

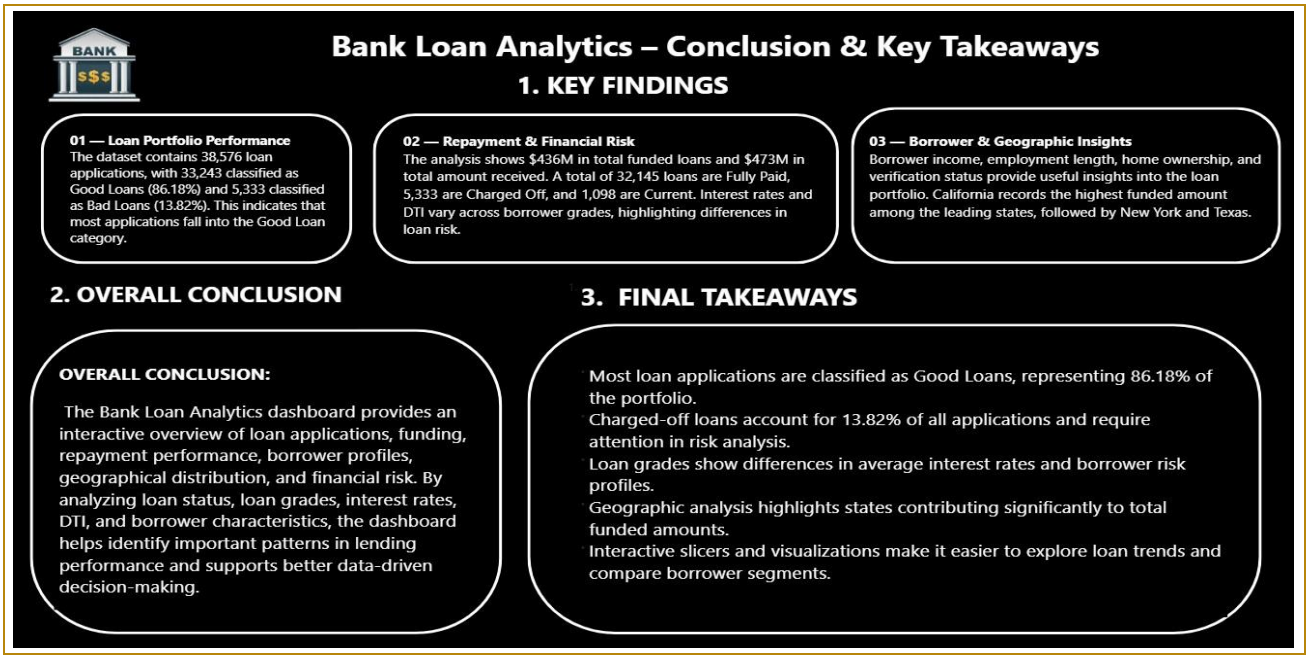


Figure 7: Conclusion & Key Takeaways page

A summary slide that presents the key findings (portfolio performance, repayment and financial risk, borrower and geographic insights), the overall conclusion and the final takeaways of the project.

5. Key Findings

Metric	Value
Total loan applications	38,576
Good Loan applications	33,243 (86.18%)
Bad Loan applications	5,333 (13.82%)
Total funded amount	$435.8M
Total amount received	$473.1M
Average loan amount	$11.3K
Average annual income	$69.6K
Average installment	$326.86
Average interest rate	12.05%
Average DTI	13.33%

- **Portfolio composition:** Out of 38,576 applications, 86.18% are Good Loans and 13.82% are Bad Loans.
- **Repayment status:** 32,145 loans are Fully Paid, 5,333 are Charged Off and 1,098 are Current.
- **Funding and collections:** Total amount received ($473.1M) exceeds total funded amount ($435.8M), which is consistent with repayments including interest.
- **Risk indicators:** Average interest rate rises from 7.35% (Grade A) to 21.40% (Grade G), while DTI varies only slightly by grade.
- **Loan purpose:** Debt consolidation is the largest purpose, both in funded amount ($232.46M) and in Charged Off loans (2.7K).
- **Geography:** California has the highest funded amount, followed by New York and Texas. California also records the most Bad Loans (1,057).
- **Borrowers:** Renters and mortgage holders make up most borrowers, and 10+ years of employment is the largest employment group.

Interpretation

The portfolio is dominated by Good Loans, but the charged-off group remains an important risk segment to monitor. Differences in interest rate and DTI by grade, together with borrower profile and state-level patterns, provide useful dimensions for further credit and portfolio analysis.

6. Conclusion and Recommendations

6.1 Conclusion

The Bank Loan Analytics dashboard provides an interactive overview of loan applications, funding, repayment performance, borrower profiles, geographical distribution and financial risk. By analysing loan status, grades, interest rates, DTI and borrower characteristics, it helps identify important patterns in lending performance and supports data-driven decision-making.

6.2 Final takeaways

- 86.18% of applications are Good Loans, indicating a strong proportion of successful accounts.
- 13.82% are Bad Loans, highlighting the importance of effective credit risk assessment.
- Total funded amount is $436M and total amount received is $473M.
- Grades show clear variation in average interest rate and DTI.

- Income, employment length, home ownership and verification status give useful insight into lending patterns.
- California, New York and Texas lead in funded amount.

6.3 Recommendations

- Monitor the Charged Off segment regularly by grade, purpose, term and state.
- Use income, DTI, employment length and verification status as comparison dimensions, not as standalone credit decisions.
- Review the highest-volume purposes and states to see where funding and charged-off loans concentrate.
- Keep KPI definitions consistent across pages and validate totals after applying slicers.
- Refresh the report whenever an updated, quality-checked dataset is available.

7. Limitations and Future Scope

7.1 Limitations

- The report describes patterns in the supplied dataset; it does not establish causation or predict whether an individual borrower will repay.
- The four missing issue dates limit date-based analysis for those records.
- The monthly trend chart on the Executive Summary page concentrates applications in a few months and omits some months from the axis, so the ISSUE_DATE field and month sorting should be re-verified before drawing time-based conclusions.
- State-level and borrower-level comparisons are descriptive summaries, not proof of why loan outcomes occurred.

7.2 Future scope

- Build a predictive model (for example logistic regression or tree-based methods) to estimate the likelihood of charge-off.
- Connect the dashboard to a live database such as PostgreSQL for automatic refresh.
- Add year-over-year and cohort analysis once the date field is fully validated.
- Publish the report to Power BI Service with role-based access for different stakeholders.

8. Conclusion

The Bank Loan Analytics Dashboard project was successfully developed using Power BI to analyze loan application data and understand the overall performance, repayment status, borrower characteristics, and financial risk associated with loan distribution. The project involved working with a cleaned financial loan dataset containing 38,576 loan records and transforming the available data into meaningful insights through interactive visualizations, key performance indicators (KPIs), and analytical reports.

The analysis provides an overview of the bank's loan portfolio, including total loan applications, total funded amount, total amount received, average loan amount, average interest rate, and average debt-to-income (DTI) ratio. The results show that approximately 86.18% of loan applications are classified as Good Loans, while 13.82% are classified as Bad Loans. This classification helps provide a clearer understanding of loan quality and highlights the importance of monitoring accounts that may involve repayment difficulties.

The financial analysis indicates that the total funded loan amount is approximately $436 million, while the total amount received is approximately $473 million. The dashboard also examines loan statuses, including Fully Paid, Current, and Charged Off, helping users understand repayment outcomes. Analysis across loan grades and sub-grades provides additional insights into variations in interest rates and borrower risk levels.

Furthermore, borrower profile analysis explores factors such as annual income, employment length, home ownership, loan purpose, and verification status. These factors help identify patterns in borrowing behaviour and provide a better understanding of the characteristics of different borrower segments. Geographical analysis enables comparisons of loan applications and funded amounts across states, offering insights into the distribution of lending activities across different regions.

The project also demonstrates the effective use of Power BI features, including DAX measures, KPI cards, slicers, filters, interactive charts, and dashboard formatting. The implementation of interactive filters allows users to explore the dataset from different perspectives, compare loan categories, examine financial performance, and identify patterns without manually analysing individual records.

Overall, this project demonstrates how data analytics and business intelligence tools can convert raw financial data into meaningful and actionable information. The developed dashboard provides a consolidated view of loan portfolio performance and supports better understanding of lending patterns, repayment behaviour, and potential credit risk areas.

As a future enhancement, the project can be extended by incorporating predictive machine learning models for loan default prediction, borrower risk classification, and repayment forecasting. Additional real-time data integration and advanced risk-scoring techniques could further improve the analytical capabilities of the dashboard.

In conclusion, the Bank Loan Analytics project successfully achieves its objective of analysing loan portfolio data through an interactive Power BI dashboard. It demonstrates the practical application of data cleaning, data analysis, DAX calculations, and data visualization to support informed, data-driven financial decision-making.